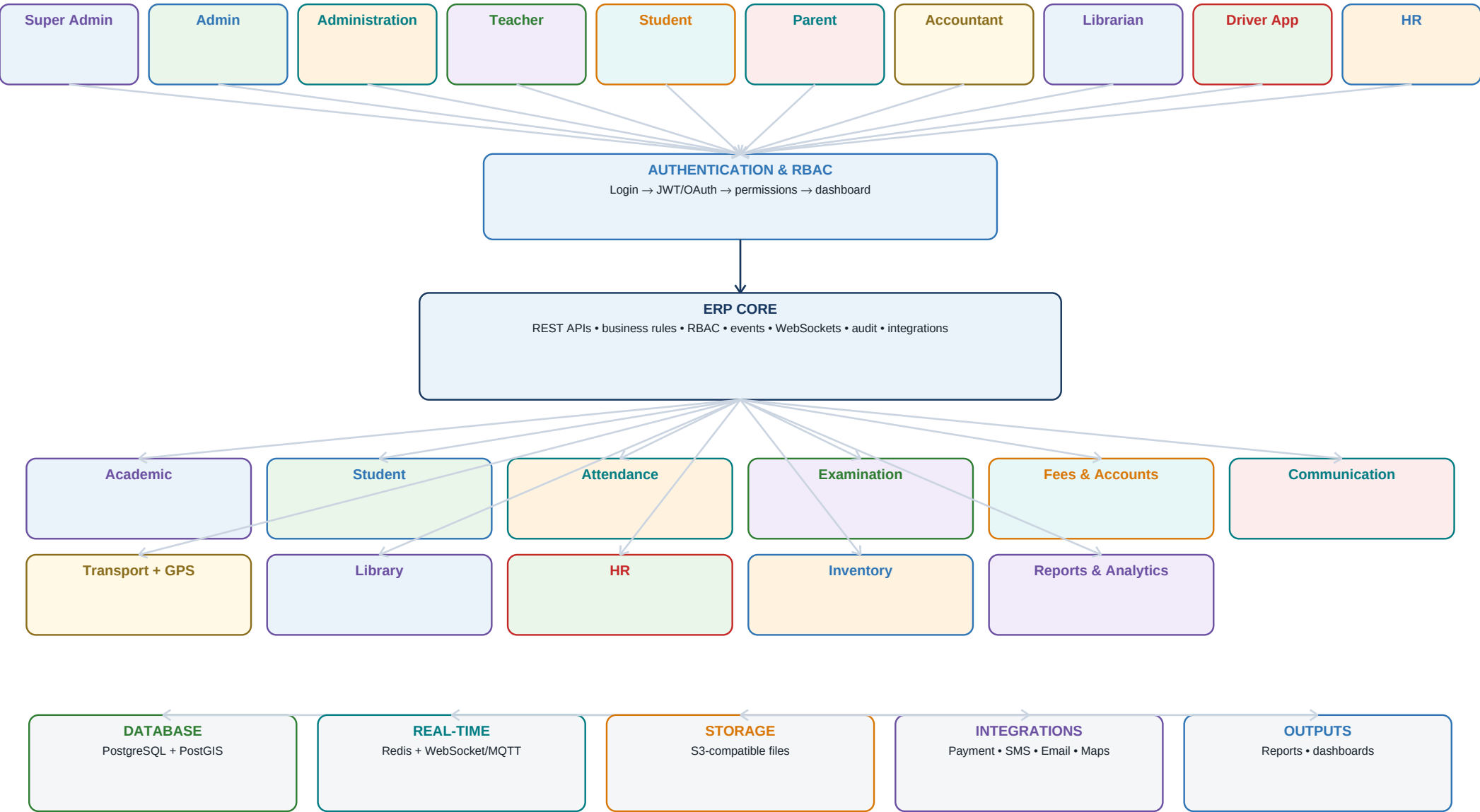


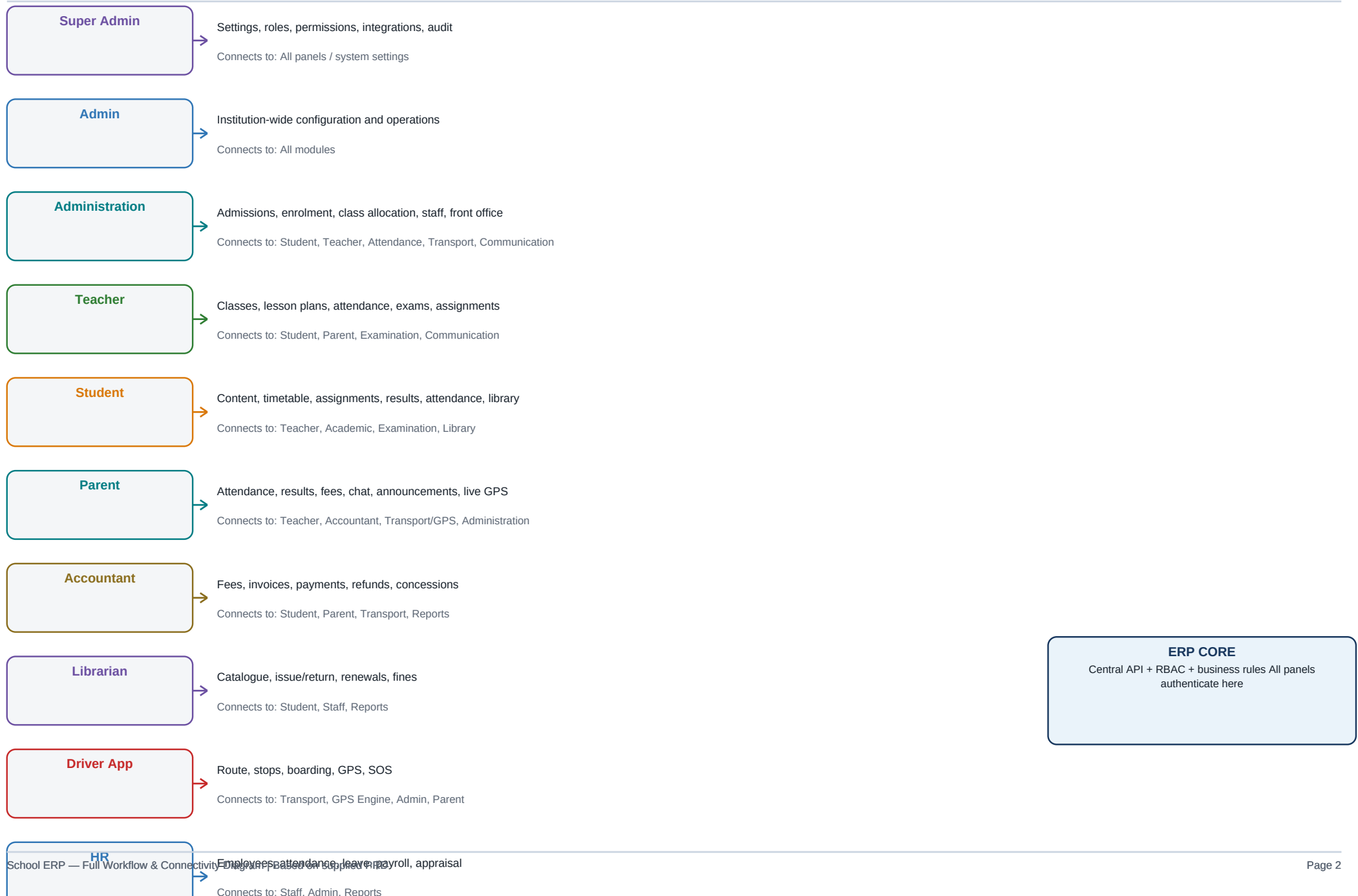
1. MASTER ERP ARCHITECTURE

Users → Authentication/RBAC → ERP Core → Functional Modules → Data / Events → Reports & Apps



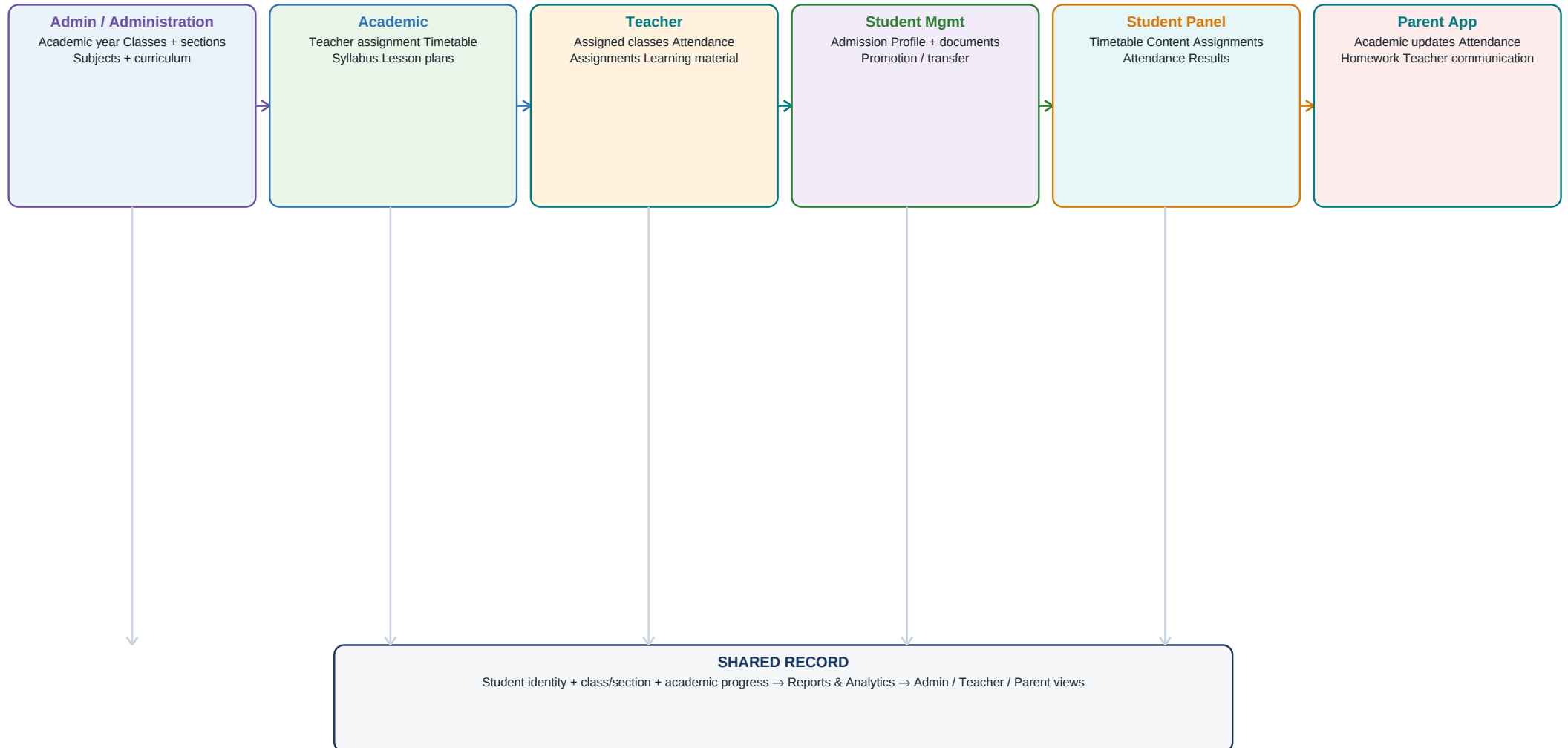
2. PANEL-TO-PANEL CONNECTIVITY

Each panel works through the ERP Core; the right side shows its main connected panels/modules



3. ACADEMIC + STUDENT WORKFLOW

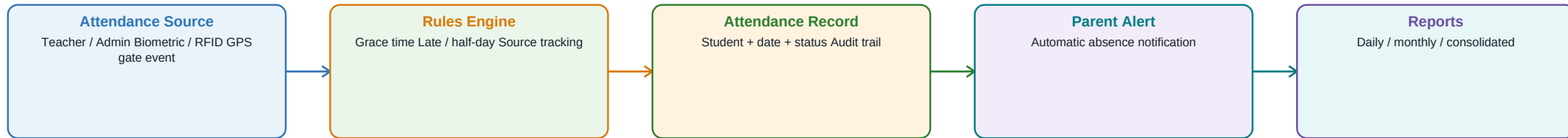
Academic setup creates the structure that teachers and students use every day



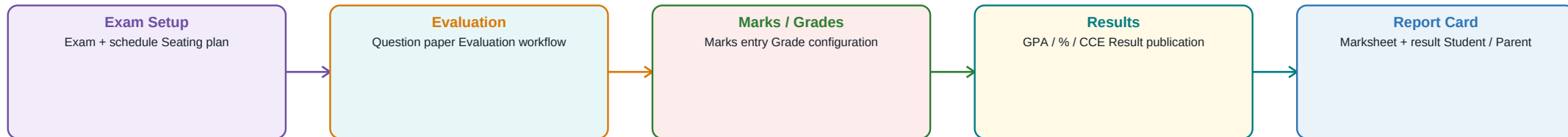
4. ATTENDANCE + EXAMINATION WORKFLOW

Attendance and examination are linked to student identity and ultimately feed reports and parent/student views

ATTENDANCE FLOW



EXAMINATION FLOW

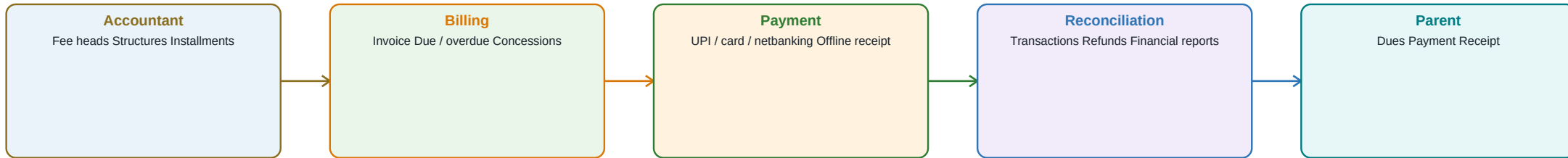


CONNECTED OUTPUT

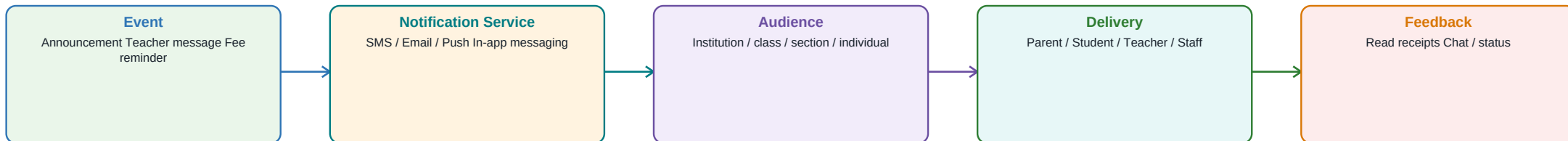
Attendance + results → Reports & Analytics → Admin / Teacher / Student / Parent

5. FEES + COMMUNICATION WORKFLOW

Finance actions generate receipts/reminders; communication routes messages to the correct audience



COMMUNICATION FLOW

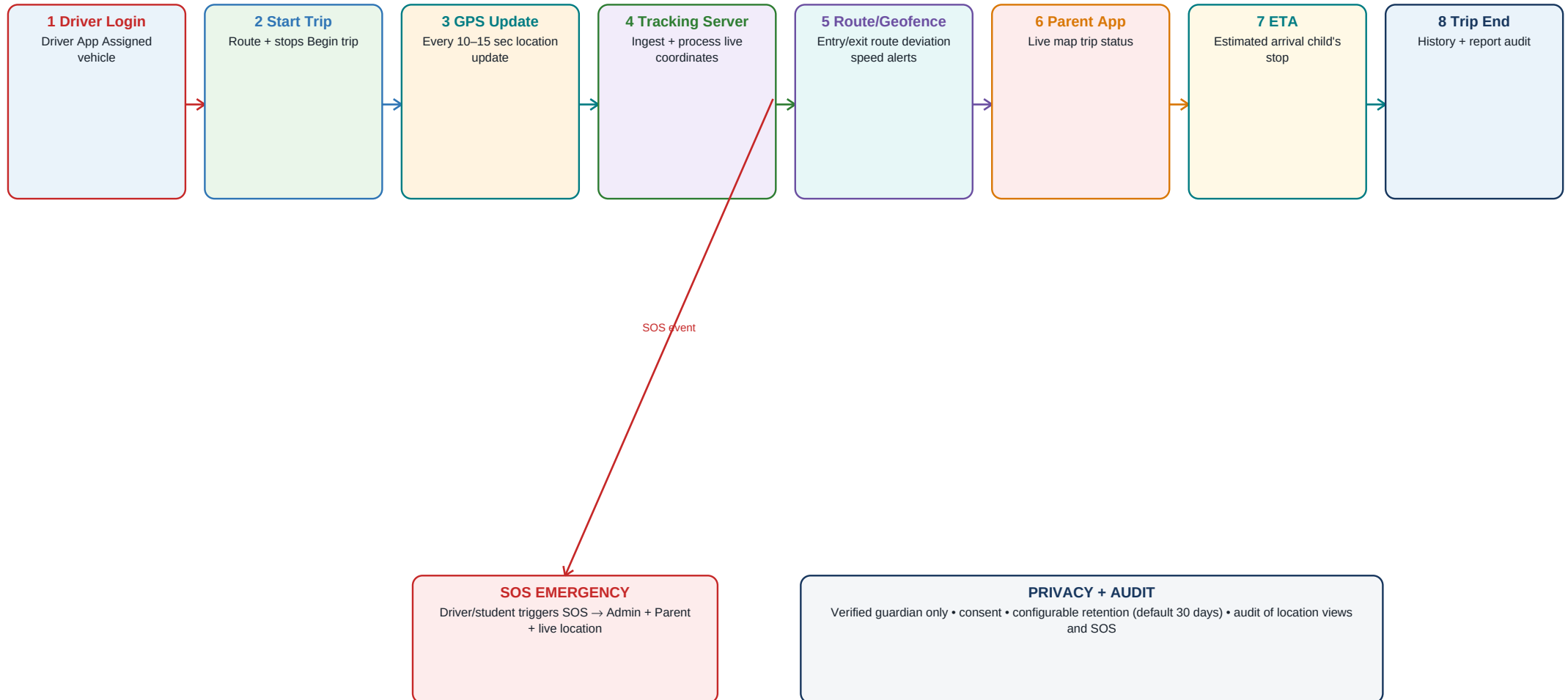


CROSS-MODULE EVENTS

Attendance → absence alert | Fees → due reminder | Teacher → parent chat | Admin → announcements

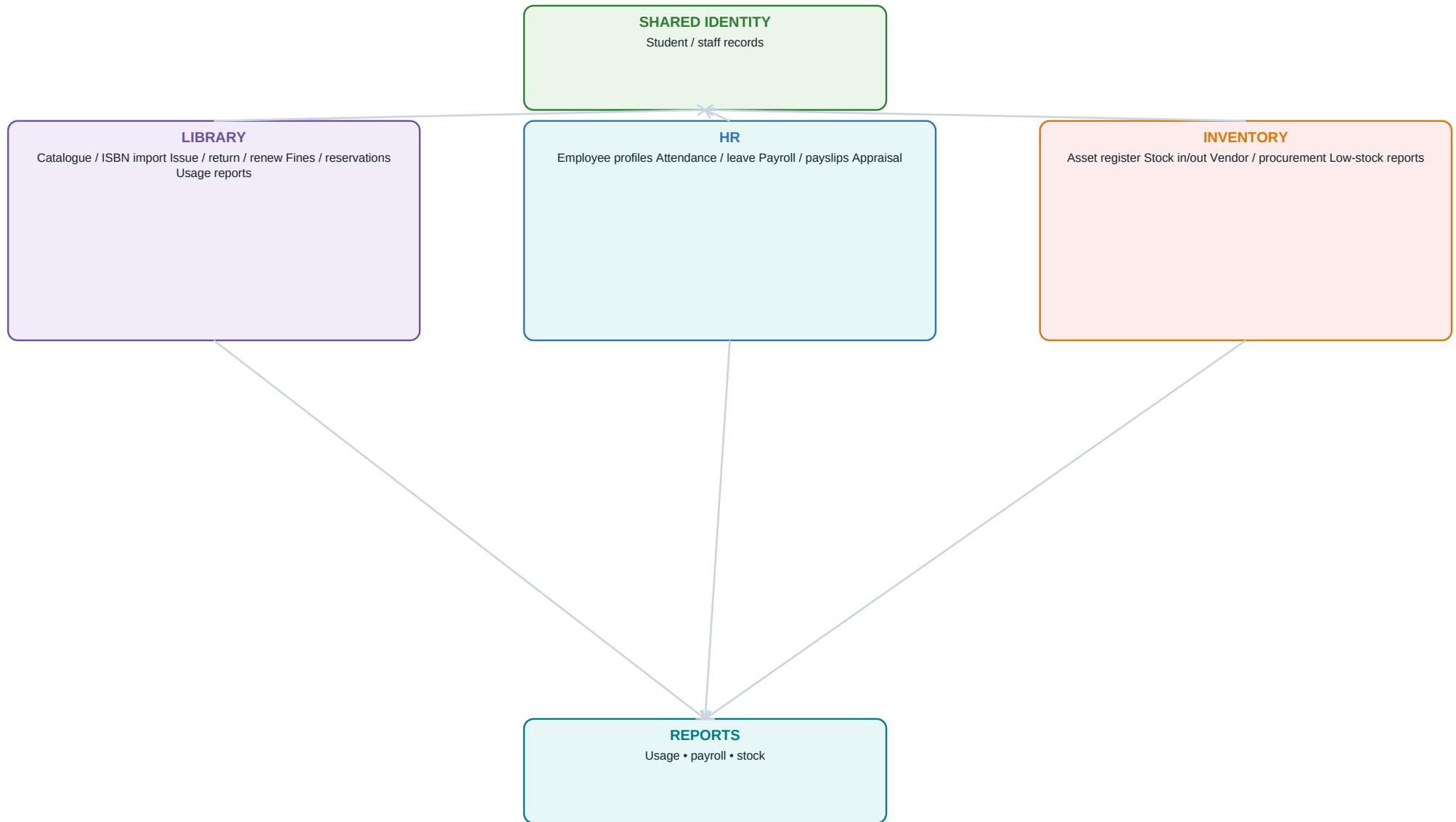
6. LIVE GPS + PARENT SAFETY WORKFLOW

Driver/device → tracking server → route/geofence engine → parent app; SOS is an emergency event path



7. LIBRARY + HR + INVENTORY WORKFLOW

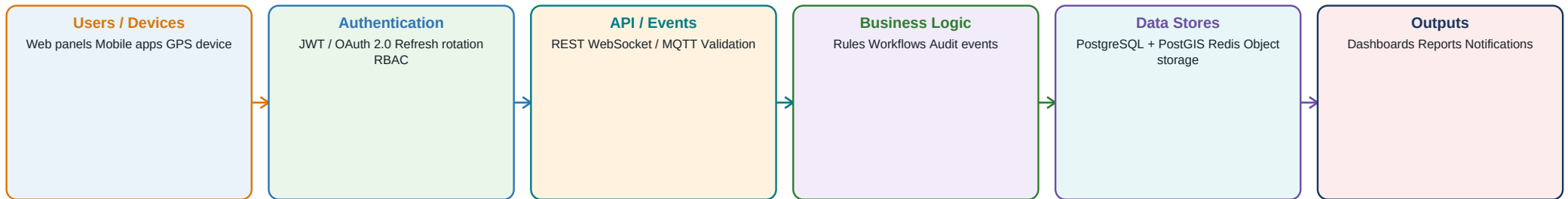
Supporting operational modules connect to student/staff records and reporting



The supplied PRD defines these modules at a functional level; exact approval chains and UI sequences are left for detailed design.

8. DATA, API + SECURITY WORKFLOW

All web/mobile/device inputs pass through authentication, API/event processing and business rules before persistence/output

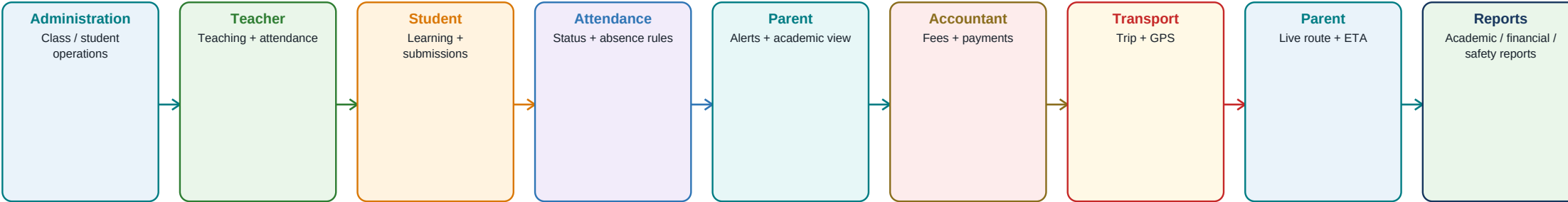


CROSS-CUTTING SECURITY

TLS in transit • AES-256 at rest • bcrypt • OWASP Top 10 protections • strict GPS authorization • audit logging • configurable retention • backups / disaster recovery

9. END-TO-END DAILY SCHOOL WORKFLOW

A single day can create linked records across academics, attendance, communication, fees and transport



← continuous data feedback

10. CROSS-MODULE EVENT MAP

Key hand-offs that make the ERP a unified system

EVENT / TRIGGER	SOURCE MODULE	PROCESS / HAND-OFF	RECEIVERS
Admission completed	Student Management	Academic → class/section allocation	Student + Parent
Teacher marks attendance	Attendance	Parent absence notification	Parent
Exam result published	Examination	Report card + analytics	Student + Parent + Teacher
Fee invoice generated	Fees	Due reminder / payment status	Parent + Accountant
Payment completed	Fees / Payment	Receipt + reconciliation	Parent + Accountant + Reports
Teacher sends message	Communication	Audience targeting	Parent / Student
Driver starts trip	Transport	GPS tracking begins	Parent + Admin
Geofence event	GPS Engine	Entry/exit alert	Parent + Admin
SOS triggered	GPS / Safety	Emergency alert + live location	Parent + Admin
Book returned	Library	Fine / inventory update	Student + Librarian
Staff leave approved	HR	Leave / attendance state	HR + Admin + Reports

11. HOW TO USE THIS DIAGRAM FOR DEVELOPMENT

Recommended reading order for the development team

1. Start with ERP Core

Implement authentication, RBAC, common user/student/staff identity, APIs, audit and shared data contracts first.

2. Build foundational modules

Academic + Student Management create the core entities used by attendance, examination, fees, library and transport.

3. Add operational workflows

Attendance, Examination, Fees, Communication, Library, HR and Inventory write records into the shared data layer.

4. Build Parent App as an aggregator

Parent consumes verified data from attendance, academics, fees, communication and transport/GPS.

5. Build GPS as a real-time subsystem

Driver App/device → tracking ingestion → route/geofence engine → Parent/Admin events; SOS follows the same event path.

SOURCE / SCOPE NOTE

This diagram is derived from the supplied School ERP PRD. It preserves the PRD's panel responsibilities, modules, connectivity, GPS flow, technology/data architecture and security requirements. Where the PRD does not define an exact business rule or UI sequence, the diagram stays at workflow level rather than inventing one.

7. Apply security and QA everywhere